

MLFF Architecture Revision 47

PERF-P2 lazy TARGET-DATA2C authority v2

mdstats project

2026-08-15

Revision 47

Revision 47 closes `PERF-P2` in `mdstats 0.20.181a0` and makes `TargetDataLadderPlan.v2` the current TARGET-DATA2C authority.

The ladder now materializes configured rungs in ascending order and stops only after the active TARGET-DATA2D shortlist width has been satisfied by the same global coverage-plus-mandatory predicate used by Stage A. The canonical width is four, but noncanonical configured widths are preserved exactly.

`v2` records the configured sequence, exact materialized prefix, global rung qualification records, early-stop qualifiers, intentionally skipped sizes, pool-unavailable sizes, monotonicity contract, last materialized rung, and stop reason. Legacy `v1` remains readable as a qualification oracle but is stale for current campaign authority.

Qualification against exhaustive `v1` preserves Stage-A survivor sizes, survivor membership, coverage-report digests, mandatory status, and the TARGET-DATA2D shortlist. A 37,633-frame supplied-data-derived forced early-stop fixture reduces median wall time from 7.867 s to 1.556 s and serialized ladder authority from 4,729,481 B to 591,058 B. The exhaustive fallback path remains exact; its noisy timing is not promoted as a speed claim.

`PERF-P3` - CPU structural and reduction hardening - is next.